

Dynamic Programming

1 Mathematical modeling of the environment

1.1 Markov chain as a model of the environment

We first define the Markov chain associated with a given environment on which we want to work. We consider $\mathbb{S}$ the finite set of states in which the agent can find itself in the environment. This set is therefore also the set of states of the Markov chain associated with the considered environment. We thus benefit from the **Markov property**:

$$\forall t \in \mathbb{N}, \forall (s_0, \dots, s_{t+1}) \in \mathbb{S}^{t+1}, P(S_{t+1} = s_{t+1} | S_t = s_t, \dots, S_0 = s_0) = P(S_{t+1} = s_{t+1} | S_t = s_t)$$

Thus, knowledge of the current state is sufficient to fully predict the future. Future states therefore depend only on the current present state.

1.2 Markov Decision Process (MDP)

We add actions and rewards to the states of the Markov chain. We denote $\mathbb{A}$ the set of actions executable by the agent. It follows that future states depend on the current state S_t and also on the chosen action A_t (which is, like S_t , a random variable). As for rewards, they are determined by S_t and A_t . We define the reward function R :

$$R : \begin{cases} \mathbb{S} \times \mathbb{A} & \rightarrow & \mathbb{R} \\ (s, a) & \mapsto & R(s, a) \end{cases}$$

where $R(s, a)$ is a quantity intrinsic to the environment. Thus, if the agent is in state s_0 and executes action a_0 , it will receive at the next time step the reward $R(s_0, a_0)$ (fixed, known in advance).

The sets $\mathbb{S}$, $\mathbb{A}$, the transition probabilities between each state, and the map R together define a Markov Decision Process (MDP). It allows us to model a given environment.

1.3 Solving an MDP

We can now formally describe the goal pursued by an agent in a given environment: that of solving the MDP associated with the environment.

We define the random variable of the **return** G_t at time t by:

$$G_t = \sum_{k=0}^h \gamma^k R_{t+k+1}$$

where $\gamma \in [0, 1[$ is the discount factor and h is the horizon. γ describes how much we wish to favor rewards close in time relative to those further in the future. The horizon h is the maximum time step. These two quantities are associated with a given environment.

To describe an agent, we define its policy, which associates to each pair (s, a) the probability of choosing action a , given that we are in state s . We therefore have:

$$\pi : \begin{cases} \mathbb{S} \times \mathbb{A} & \rightarrow & [0, 1] \\ (s, a) & \mapsto & \pi(a|s) \end{cases}$$

For π to be well-defined as a policy, we require:

$$\forall s \in \mathbb{S}, \sum_{a \in \mathbb{A}} \pi(a|s) = 1$$

We say that the agent **follows** policy π .

We can finally formalize our objective: solving an MDP means finding a policy π_* that maximizes the return G_t for all t . In the context of this document, we consider the reward function R and the transition probabilities to be known:

$$\forall t \in \mathbb{N}, \forall (s, a, s') \in \mathbb{S} \times \mathbb{A} \times \mathbb{S}, P(S_{t+1} = s' \mid S_t = s, A_t = a) \text{ is known}$$

For brevity, we write $P(s'|s, a)$ to denote the quantity $P(S_{t+1} = s' \mid S_t = s, A_t = a)$.

2 First problem: evaluating a policy

With the goal of finding a policy π_* that maximizes the return G_t for all t , we will work iteratively. We start from an initial policy (in practice, a naive one) and **improve** it at each iteration. To ensure that an improvement takes place, we must be able to evaluate a given policy. We can then evaluate the first policy and the one resulting from the first iteration, and verify that the latter is indeed better than the former. To this end, we define the **value function** v_π associated with policy π by:

$$\forall t \in \mathbb{N}, \forall s \in \mathbb{S}, v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$$

where $\mathbb{E}_\pi$ denotes an expectation, indexed by π to emphasize that it is the expected return obtained **by following policy** π . We view v_π as an element of $\mathbb{R}^{\text{card}(\mathbb{S})}$. This definition can be expanded using knowledge of the transition probabilities and rewards of the environment:

$$\begin{aligned} \forall t \in \mathbb{N}, \forall s \in \mathbb{S}, v_\pi(s) &= \mathbb{E}_\pi[R_{t+1} + \gamma G_{t+1} \mid S_t = s] \quad (\text{decomposition of } G_t \text{ via Chasles}) \\ &= \mathbb{E}_\pi[R_{t+1} \mid S_t = s] + \gamma \mathbb{E}_\pi[G_{t+1} \mid S_t = s] \quad (\text{linearity of expectation}) \\ &= \sum_{a \in \mathbb{A}} \pi(a|s) R(s, a) + \gamma \sum_{a \in \mathbb{A}} \pi(a|s) \sum_{s' \in \mathbb{S}} P(s'|s, a) v_\pi(s') \end{aligned}$$

Indeed, for the first sum, the reward R_{t+1} depends only on the state and the action taken at time t . We therefore compute its expectation as a function of the chosen action. As for the second sum, we look at the future as a function of the action taken at time t . For each action, there is a certain probability $P(s'|s, a)$ of landing in state s' (where s' ranges over $\mathbb{S}$). From there, we know the value function at that state: $v_\pi(s')$.

This equation on v_π is called the **Bellman equation**. It yields a linear system of $\text{card}(\mathbb{S})$ equations in $\text{card}(\mathbb{S})$ unknowns, which can be solved analytically to compute v_π . However, this approach does not generalize to what follows. We therefore choose to solve it iteratively with the following algorithm.

We construct an algorithm, the POLICY EVALUATION algorithm, to compute v_π for a given policy π . This algorithm starts from an arbitrary estimate of v_π and iteratively applies an operator L_π defined by:

$$L_\pi : \begin{cases} \mathbb{R}^{\text{card}(\mathbb{S})} & \rightarrow \mathbb{R}^{\text{card}(\mathbb{S})} \\ V & \mapsto R_\pi + \gamma T_\pi V \end{cases}$$

where R_π is the column vector of length $\text{card}(\mathbb{S})$ that contains, for each state s , the average reward obtained with π when leaving that state:

$$\sum_{a \in \mathbb{A}} \pi(a|s) R(s, a)$$

and T_π is the one-step transition matrix of policy π . Writing $\mathbb{S} = \{s_1, \dots, s_N\}$, we have for T_π :

$$\forall (i, j) \in \llbracket 1, N \rrbracket^2, (T_\pi)_{i,j} = P(S_{t+1} = s_j | S_t = s_i)$$

Given π , we can compute R_π and T_π from our knowledge of the environment dynamics.

It is worth noting that $L_\pi(V)$ is a matrix rewriting of the right-hand side of the equality defining $v_\pi(s)$.

The goal of the next paragraph is to use a fixed-point theorem to show that iterating L_π guarantees convergence of the POLICY EVALUATION algorithm.

A value function will therefore be represented as an element of $\mathbb{R}^{\text{card}(\mathbb{S})}$. For $s \in \mathbb{S}$, we write $V(s)$ for the value of state s under V (we use function notation since V resembles the idea of a function).

2.1 Fixed-point theorem on $\mathbb{R}^n$

We fix $n \in \mathbb{N}^*$ throughout this paragraph.

Definition 1 *Infinity norm on $\mathbb{R}^n$*

For $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, we define the **infinity norm** of x , denoted $\|x\|_\infty$, by:

$$\|x\|_\infty = \max_{i \in [1, n]} |x_i|$$

Definition 2 *Sequences and convergence in $\mathbb{R}^n$*

Consider $(u_n) \in (\mathbb{R}^n)^\mathbb{N}$, a sequence of elements of $\mathbb{R}^n$, and $l \in \mathbb{R}^n$. We say that u converges to l if:

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N, \|u_n - l\|_\infty < \varepsilon$$

Definition 3 *Cauchy sequence in $\mathbb{R}^n$*

Consider $(u_n) \in (\mathbb{R}^n)^\mathbb{N}$. We say that u is **Cauchy** for the infinity norm if:

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall p \geq N, \forall q \geq N, \|u_p - u_q\|_\infty < \varepsilon$$

in other words, u is Cauchy if all terms of the sequence are as close as desired under the infinity distance (the distance associated with the infinity norm), from some rank onward.

Proposition 1 *Every Cauchy sequence in $\mathbb{R}$ converges in $\mathbb{R}$.*

Proof

Let $(u_n) \in \mathbb{R}^\mathbb{N}$ be a Cauchy sequence.

We first show that (u_n) is bounded. We apply the Cauchy definition for (u_n) with $\varepsilon = 1$.

There exists $N_1 \in \mathbb{N}$ such that:

$$\forall p \geq N_1, \forall q \geq N_1, |u_p - u_q| \leq 1 \quad (\text{on } \mathbb{R}, \|\cdot\|_\infty = |\cdot|)$$

Taking $n \geq N_1$, we get:

$$|u_n - u_{N_1}| \leq 1$$

By the triangle inequality for $|\cdot|$:

$$|u_n| - |u_{N_1}| \leq |u_n - u_{N_1}| \leq 1$$

hence:

$$|u_n| \leq 1 + |u_{N_1}|$$

Setting $d = 1 + |u_{N_1}|$ and

$$M = \max(|u_1|, \dots, |u_{N_1-1}|, d)$$

it follows that:

$$\forall n \in \mathbb{N}, |u_n| \leq M$$

which shows that (u_n) is bounded.

We now show that (u_n) converges. By the Bolzano–Weierstrass theorem, we can extract a convergent subsequence from u . There exists $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ strictly increasing such that $(u_{\varphi(n)})$ converges to some $l \in \mathbb{R}$. We show that u converges to l .

Let $\varepsilon > 0$. By convergence of $(u_{\varphi(n)})$ with $\frac{\varepsilon}{2}$, there exists $N_0 \in \mathbb{N}$ such that:

$$\forall n \geq N_0, |u_{\varphi(n)} - l| \leq \frac{\varepsilon}{2}$$

By the Cauchy property for (u_n) with $\frac{\varepsilon}{2}$, there exists $N_1 \in \mathbb{N}$ such that:

$$\forall p \geq N_1, \forall q \geq N_1, |u_p - u_q| \leq \frac{\varepsilon}{2}$$

Let $n \geq \max(N_0, N_1)$. We have:

$$\begin{aligned} |u_n - l| &= |u_n - u_{\varphi(n)} + u_{\varphi(n)} - l| \\ &\leq |u_n - u_{\varphi(n)}| + |u_{\varphi(n)} - l| \quad (\text{triangle inequality}) \\ &\leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \quad (\text{since } \varphi(n) \geq n) \\ &= \varepsilon \end{aligned}$$

which shows that u converges to l .

Proposition 2 *Every Cauchy sequence in $\mathbb{R}^n$ converges in $\mathbb{R}^n$.*

Proof

We prove the result by induction on $n \in \mathbb{N}^*$.

- $\mathbb{R}^1$ satisfies the property **at rank 1** (cf. Prop. 4).
- **Assume the property holds** for some $n \in \mathbb{N}^*$.

We show that $\mathbb{R}^{n+1} = \mathbb{R}^n \times \mathbb{R}$ also satisfies the property.

We denote by $|\cdot|$, $\|\cdot\|_{n,\infty}$, $\|\cdot\|_{n+1,\infty}$ the infinity norms on $\mathbb{R}$, $\mathbb{R}^n$, $\mathbb{R}^{n+1}$ respectively. (On $\mathbb{R}$, this is the absolute value $|\cdot|$.) We can rewrite the last norm in terms of the first two: for $x = (\lambda, x_1) \in \mathbb{R}^{n+1}$ (with $x_1 \in \mathbb{R}$):

$$\|x\|_{n+1,\infty} = \max\left(\|\lambda\|_{n,\infty}, |x_1|\right) \quad (*)$$

Let $(u_n) \in (\mathbb{R}^{n+1})^{\mathbb{N}}$ be a Cauchy sequence.

There exist $(v_n) \in (\mathbb{R}^n)^{\mathbb{N}}$ and $(w_n) \in \mathbb{R}^{\mathbb{N}}$ such that:

$$(u_n) = ((v_n, w_n))$$

For any $(p, q) \in \mathbb{N}^2$, using $(*)$:

$$\|v_p - v_q\|_{n,\infty} \leq \|u_p - u_q\|_{n+1,\infty}$$

This shows that the distance between terms of (v_n) is no greater than the distance between corresponding terms of (u_n) . Since (u_n) is Cauchy, (v_n) is also Cauchy in $\mathbb{R}^n$ under $\|\cdot\|_{n,\infty}$. The same argument shows that (w_n) is Cauchy in $\mathbb{R}$ under $|\cdot|$.

By the induction hypothesis applied to $\mathbb{R}^n$ and $\mathbb{R}$, there exist $l_v \in \mathbb{R}^n$ and $l_w \in \mathbb{R}$ such that $(v_n) \rightarrow l_v$ and $(w_n) \rightarrow l_w$.

Set $l_u = (l_v, l_w)$. We show that (u_n) converges to l_u .

Let $\varepsilon > 0$. By convergence of (v_n) and (w_n) , there exist integers N_0, N_1 such that:

$$\forall n \geq N_0, \|v_n - l_v\|_{n,\infty} < \varepsilon$$

and

$$\forall n \geq N_1, |w_n - l_w| < \varepsilon$$

Setting $N = \max(N_0, N_1)$ and taking $n \geq N$:

$$\|v_n - l_v\|_{n,\infty} < \varepsilon \quad \text{and} \quad |w_n - l_w| < \varepsilon$$

hence:

$$\max\left(\|v_n - l_v\|_{n,\infty}, |w_n - l_w|\right) = \|u_n - l_u\|_{n+1,\infty} < \varepsilon$$

which proves that (u_n) converges in $\mathbb{R}^{n+1}$. The property holds at rank $n + 1$.

- **Conclusion:** for all $n \in \mathbb{N}^*$, every Cauchy sequence in $\mathbb{R}^n$ converges in $\mathbb{R}^n$.

Definition 4 Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $a \in \mathbb{R}^n$. We say that f is **continuous at a** if:

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R}^n \quad \|x - a\|_\infty < \delta \implies \|f(x) - f(a)\|_\infty < \varepsilon$$

We say that f is **continuous** if it is continuous at every $a \in \mathbb{R}^n$.

Proposition 3 Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be contractive for the infinity norm. Then f is continuous.

Proof

For a given $\varepsilon > 0$, one can take $\delta = \frac{\varepsilon}{k}$, where f is k -Lipschitz.

Proposition 4 *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be continuous and let u be a sequence in $\mathbb{R}^n$ converging to $a \in \mathbb{R}^n$. Then the sequence $(f(u_n))$ converges to $f(a)$.*

Proof

Apply the definition of continuity of f and then that of convergence of u .

Proposition 5 *A fixed-point theorem on $\mathbb{R}^n$.*

Let f be a contractive map from $\mathbb{R}^n$ to $\mathbb{R}^n$. Then f has a unique fixed point α . Moreover, if we define the sequence u by:

$$\begin{cases} u_0 = x_0 \in \mathbb{R}^n \\ u_{n+1} = f(u_n), \forall n \in \mathbb{N} \end{cases} \quad (1)$$

then u converges to α .

Proof

Since f is contractive for the infinity norm, there exists $k \in [0, 1[$ such that f is k -Lipschitz.

Uniqueness. Suppose α and β are two fixed points of f . Since f is k -Lipschitz:

$$\|\alpha - \beta\|_\infty = \|f(\alpha) - f(\beta)\|_\infty \leq k\|\alpha - \beta\|_\infty$$

hence $\|\alpha - \beta\|_\infty = 0$, so $\alpha = \beta$.

Existence. Since $f(\mathbb{R}^n) \subset \mathbb{R}^n$, the sequence u is well-defined. For $n \in \mathbb{N}^*$:

$$\|u_{n+1} - u_n\|_\infty = \|f(u_n) - f(u_{n-1})\|_\infty \leq k\|u_n - u_{n-1}\|_\infty$$

and therefore, by induction on $n \in \mathbb{N}$:

$$\|u_{n+1} - u_n\|_\infty \leq k^n \|u_1 - u_0\|_\infty \quad (*)$$

We now show that u is Cauchy. Let $(n, l) \in (\mathbb{N}^*)^2$. We have:

$$\begin{aligned} \|u_{n+l} - u_n\|_\infty &\leq \sum_{i=0}^{l-1} \|u_{n+i+1} - u_{n+i}\|_\infty \quad (\text{triangle inequality applied } l-1 \text{ times}) \\ &\leq \|u_1 - u_0\|_\infty \sum_{i=0}^{l-1} k^{n+i} \quad (\text{using } (*)) \\ &\leq \|u_1 - u_0\|_\infty \frac{k^n}{1-k} \quad (k \in [0, 1]) \end{aligned}$$

thus, for large enough n , the distance between u_{n+l} and u_n tends to 0: we conclude that u is Cauchy. Since it is a sequence in $\mathbb{R}^n$, we know by Prop. 9.1.2 that u converges in $\mathbb{R}^n$. Denote its limit α . For all $n \in \mathbb{N}$:

$$u_{n+1} = f(u_n)$$

Since f is contractive, it is in particular continuous (cf. Prop. 9.1.3). Passing to the limit:

$$\alpha = f(\alpha)$$

which proves the existence of a fixed point for f .

2.2 Back to the problem: convergence of the policy evaluation algorithm

We define the POLICY EVALUATION algorithm as follows:

Algorithm 1: Policy Evaluation

```

1 Input:  $\pi$ 
2 Initialize  $V(s) = 0$  for each  $s \in \mathbb{S}$ 
3
4 repeat
5   |   foreach  $s$  do
6   |   |    $V(s) \leftarrow \sum_a \pi(a|s)R(s, a) + \gamma \sum_a \pi(a|s) \sum_{s'} P(s'|s, a)V(s')$ 
7   |   end
8 until  $V$  has almost stopped changing
9
10 return  $V \approx v_\pi$ 

```

Proposition 6 L_π is contractive for the infinity norm on $\mathbb{R}^{\text{card}(\mathbb{S})}$, i.e. there exists $k \in [0, 1[$ such that

$$\forall (V_1, V_2) \in (\mathbb{R}^{\text{card}(\mathbb{S})})^2, \left\| L_\pi(V_1) - L_\pi(V_2) \right\|_\infty \leq k \left\| V_1 - V_2 \right\|_\infty$$

Proof

Let $(V_1, V_2) \in (\mathbb{R}^{\text{card}(\mathbb{S})})^2$. We have:

$$\begin{aligned}
\left\| L_\pi(V_1) - L_\pi(V_2) \right\|_\infty &= \left\| R_\pi + \gamma T_\pi V_1 - (R_\pi + \gamma T_\pi V_2) \right\|_\infty \\
&= \left\| \gamma T_\pi (V_1 - V_2) \right\|_\infty \\
&\leq \gamma \left\| T_\pi \right\|'_\infty \left\| V_1 - V_2 \right\|_\infty \quad (\gamma \in [0, 1]) \\
&\leq \gamma \left\| V_1 - V_2 \right\|_\infty
\end{aligned}$$

since T_π is stochastic, its matrix infinity norm (in $\mathcal{M}_{\text{card}(\mathbb{S})}(\mathbb{R})$) is bounded above by 1.

Proposition 7 *The POLICY EVALUATION algorithm converges, to v_π .*

Proof

L_π is contractive for the infinity norm on the vector space $\mathbb{R}^{\text{card}(\mathbb{S})}$, which is a complete normed space by Proposition 5. We can therefore apply the fixed-point theorem (Proposition 6) to L_π . This map has a unique fixed point $V_* \in \mathbb{R}^{\text{card}(\mathbb{S})}$, so the algorithm terminates.

Moreover, v_π satisfies the Bellman equations, which translates to:

$$L_\pi(v_\pi) = v_\pi$$

i.e., v_π is a fixed point of L_π . By uniqueness of this fixed point:

$$V_* = v_\pi$$

which proves the correctness of the algorithm.

We can even bound the distance between the value function obtained and the target v_π (in other words, the error of the algorithm):

Proposition 8 *Suppose that during the iteration of L_π starting from V_0 , at some step $n \in \mathbb{N}^*$:*

$$\left\| L_\pi^n(V_0) - L_\pi^{n-1}(V_0) \right\|_\infty \leq \varepsilon$$

for some fixed $\varepsilon > 0$. Then we have the following error bound:

$$\left\| v_\pi - L_\pi^n(V_0) \right\|_\infty \leq \frac{\gamma \varepsilon}{1 - \gamma}$$

Proof

To simplify notation, write for all $n \in \mathbb{N}^*$:

$$u_n = L_\pi^n(V_0)$$

For $n \in \mathbb{N}^*$, since L_π is contractive for the infinity norm:

$$\begin{aligned} \left\| u_{n+1} - u_n \right\|_\infty &= \left\| L_\pi(u_n) - L_\pi(u_{n-1}) \right\|_\infty \\ &\leq \gamma \left\| u_n - u_{n-1} \right\|_\infty \end{aligned}$$

and by induction, for all $(n, p) \in \mathbb{N}^* \times \mathbb{N}$:

$$\left\| u_{n+p} - u_{n+p-1} \right\|_\infty \leq \gamma^p \left\| u_n - u_{n-1} \right\|_\infty \quad (*)$$

Let $(n, p) \in \mathbb{N}^* \times \mathbb{N}$. We have:

$$\begin{aligned}
\|u_{n+p} - u_n\|_\infty &\leq \sum_{k=1}^p \|u_{n+p-k+1} - u_{n+p-k}\|_\infty \quad (\text{telescoping} + \text{triangle inequality}) \\
&\leq \sum_{k=1}^p \left(\gamma^{p-k+1} \|u_n - u_{n-1}\|_\infty \right) \quad ((*) \text{ with } p \leftarrow p - k + 1) \\
&= \left(\sum_{k=1}^p \gamma^k \right) \|u_n - u_{n-1}\|_\infty \quad (\text{index substitution}) \\
&= \gamma \frac{1 - \gamma^p}{1 - \gamma} \|u_n - u_{n-1}\|_\infty \quad (\gamma \neq 1)
\end{aligned}$$

Letting $p \rightarrow +\infty$ in the above inequality:

$$\|v_\pi - u_n\|_\infty \leq \frac{\gamma}{1 - \gamma} \|u_n - u_{n-1}\|_\infty$$

since we know from Prop. 7 that the policy evaluation algorithm converges to v_π , and $\gamma \in [0, 1[$. Thus, if $\|u_n - u_{n-1}\|_\infty \leq \varepsilon$, the desired inequality follows:

$$\|v_\pi - L_\pi^n(V_0)\|_\infty \leq \frac{\gamma \varepsilon}{1 - \gamma}$$

We now have an algorithm capable of computing the value function of a given policy.

3 VALUE ITERATION algorithm

Still with the goal of solving an MDP, we seek the optimal policy π_* . This policy, by definition, takes in each state, in a deterministic manner, the action that maximizes the subsequent return G_t . (Thus $\pi(a|s) = 0$ except for a single action.) We first try to determine its value function v_* .

Fix $t \in \mathbb{N}$ and $s \in \mathbb{S}$. Since π_* is optimal, we have:

$$\begin{aligned} v_*(s) &= \max_a \left(\mathbb{E}_{\pi_*} [R_{t+1} + \gamma G_{t+1} | S_t = s, A_t = a] \right) \quad (\text{we recover the highest achievable value}) \\ &= \max_a \left(\mathbb{E}_{\pi_*} [R_{t+1} | S_t = s, A_t = a] + \gamma \mathbb{E}_{\pi_*} [G_{t+1} | S_t = s, A_t = a] \right) \quad (\text{linearity of expectation}) \\ &= \max_a \left(R(s, a) + \gamma \sum_{s' \in \mathbb{S}} P(s' | s, a) v_*(s') \right) \end{aligned}$$

This is known as the **Bellman optimality equation**. Just as with the Bellman equation, it yields a system of $\text{card}(\mathbb{S})$ equations satisfied by v_* . However, this system is no longer linear due to the presence of the max operator. We solve it iteratively with the VALUE ITERATION algorithm.

This algorithm, very similar to POLICY EVALUATION, starts from an arbitrary estimate of v_* and iteratively applies an operator H defined by:

$$H : \begin{cases} \mathbb{R}^{\text{card}(\mathbb{S})} & \rightarrow \mathbb{R}^{\text{card}(\mathbb{S})} \\ V & \mapsto \max_a R^a + \gamma T^a V \end{cases}$$

where R^a is the column vector of length $\text{card}(\mathbb{S})$ containing, for each state s , the value $R(s, a)$, and T^a is the one-step transition matrix for action a . For $a \in \mathbb{A}$, writing $\mathbb{S} = \{s_1, \dots, s_N\}$:

$$\forall (i, j) \in \llbracket 1, N \rrbracket^2, \quad (T^a)_{i,j} = P(s_j | s_i, a)$$

Again, it is worth noting that $H(V)$ is a rewriting of the right-hand side of the Bellman optimality equation.

Proposition 9 *H is contractive for the infinity norm, i.e. there exists $k \in [0, 1[$ such that*

$$\forall (V_1, V_2) \in (\mathbb{R}^{\text{card}(\mathbb{S})})^2, \left\| H(V_1) - H(V_2) \right\|_{\infty} \leq k \left\| V_1 - V_2 \right\|_{\infty}$$

Proof

Let $s \in \mathbb{S}$. Denote by a_s^* an optimal action in state s (not necessarily unique):

$$a_s^* = \operatorname{argmax}_a \left(R(s, a) + \gamma \sum_{s' \in \mathbb{S}} P(s'|s, a) v_*(s') \right)$$

Let $(V_1, V_2) \in (\mathbb{R}^{\text{card}(\mathbb{S})})^2$. Without loss of generality, suppose $H(V_1)(s) \geq H(V_2)(s)$. By definition of H :

$$\begin{aligned} 0 &\leq H(V_1)(s) - H(V_2)(s) \\ &= R(s, a_s^*) + \gamma \sum_{s' \in \mathbb{S}} P(s'|s, a_s^*) V_1(s') - R(s, a_s^*) - \gamma \sum_{s' \in \mathbb{S}} P(s'|s, a_s^*) V_2(s') \\ &= \gamma \sum_{s' \in \mathbb{S}} P(s'|s, a_s^*) (V_1(s') - V_2(s')) \quad (\text{linearity of sum}) \\ &\leq \gamma \sum_{s' \in \mathbb{S}} P(s'|s, a_s^*) \left\| V_1 - V_2 \right\|_{\infty} \quad (\text{bound by infinity norm}) \\ &= \gamma \left\| V_1 - V_2 \right\|_{\infty} \sum_{s' \in \mathbb{S}} P(s'|s, a_s^*) \\ &= \gamma \left\| V_1 - V_2 \right\|_{\infty} \quad (\text{probabilities sum to 1}) \end{aligned}$$

The symmetric case $H(V_2)(s) \geq H(V_1)(s)$ gives the same bound (since $\|x\|_{\infty} = \|-x\|_{\infty}$).

Since the inequality holds for all states $s \in \mathbb{S}$, it holds in particular for the state maximizing the difference, giving:

$$\left\| H(V_1) - H(V_2) \right\|_{\infty} \leq \gamma \left\| V_1 - V_2 \right\|_{\infty}$$

which shows that H is contractive for the infinity norm, with $k = \gamma$.

The VALUE ITERATION algorithm is then defined as follows:

Algorithm 2: VALUE ITERATION

```

1 Initialize  $V(s) = 0$  for each  $s \in \mathbb{S}$ 
2
3 repeat
4   foreach  $s$  do
5      $V(s) \leftarrow \max_a \left[ R(s, a) + \gamma \sum_{s'} P(s'|s, a) V(s') \right]$ 
6   end
7 until  $V$  has almost stopped changing
8
9 return  $V \approx v_*$ 

```

This consists of iterating the operator H , starting from an arbitrary value function (here, the zero value function), i.e. an element of $\mathbb{R}^{\text{card}(\mathbb{S})}$.

Proposition 10 *The VALUE ITERATION algorithm converges to v_* .*

Proof

H is contractive for the infinity norm on $\mathbb{R}^{\text{card}(\mathbb{S})}$, which is a complete normed space by Proposition 2. By the fixed-point theorem (Proposition 5) applied to H , this map has a unique fixed point $v \in \mathbb{R}^{\text{card}(\mathbb{S})}$, so the algorithm terminates. Moreover, since v_* satisfies the Bellman optimality equations:

$$H(v_*) = v_*$$

By uniqueness of the fixed point of H :

$$v = v_*$$

which proves the correctness of the algorithm.

Proposition 11 *Let V_0 be a value function and $\varepsilon > 0$. If*

$$\left\| H^{n+1}(V_0) - H^n(V_0) \right\|_{\infty} \leq \varepsilon$$

then

$$\left\| H^{n+1}(V_0) - v_* \right\|_{\infty} \leq \frac{\varepsilon}{1 - \gamma}$$

Proof

Same proof as Prop. 8 (we only used the fact that L_π is contractive, and H is as well).

We thus have an algorithm for computing the value function of π_* to a precision we can determine (cf. Prop. 11). However, we want to obtain π_* itself. We can deduce it from v_* by assigning, for each state s , probability 1 to the action a that maximizes $R(s, a) + \gamma \sum_{s'} P(s'|s, a) v_*(s')$. But we only have access to an approximation of v_* : we must therefore ensure that the value function of the policy extracted from this approximation **remains close** to v_* .

Proposition 12 *Let V_0 be a value function and $\varepsilon > 0$. If*

$$\left\| H^{n+1}(V_0) - H^n(V_0) \right\|_\infty \leq \varepsilon$$

then, denoting π_n the policy defined by

$$\forall s \in \mathbb{S}, \pi_n(s) = \operatorname{argmax}_a \left(R(s, a) + \gamma \sum_{s' \in \mathbb{S}} P(s'|s, a) H^n(V_0)(s') \right)$$

we have:

$$\left\| v_{\pi_n} - v_* \right\|_\infty \leq \frac{2\gamma\varepsilon}{1-\gamma}$$

Proof

We have:

$$\begin{aligned} \left\| v_{\pi_n} - v_* \right\|_\infty &= \left\| v_{\pi_n} - H^n(V_0) + H^n(V_0) - v_* \right\|_\infty \\ &\leq \left\| v_{\pi_n} - H^n(V_0) \right\|_\infty + \left\| H^n(V_0) - v_* \right\|_\infty \\ &\leq \frac{\gamma\varepsilon}{1-\gamma} + \frac{\gamma\varepsilon}{1-\gamma} \quad (\text{Props. 8 and 11}) \\ &= \frac{2\gamma\varepsilon}{1-\gamma} \end{aligned}$$

Prop. 8 applies because π_n was constructed using the POLICY EVALUATION operator L_{π_n} . Prop. 11 applies directly from the assumed inequality.

Proposition 13 *The time complexity of VALUE ITERATION is $\Theta\left(n|\mathbb{S}|^2|\mathbb{A}|\right)$, where n denotes the number of iterations performed, $|\mathbb{S}| = \text{card}(\mathbb{S})$, and $|\mathbb{A}| = \text{card}(\mathbb{A})$.*

Proof

We first compute the time complexity of a single iteration.

The VALUE ITERATION algorithm can be written as:

```
value_iteration:
  for each  $s \in \mathbb{S}$ :
    for each  $a \in \mathbb{A}$ :
      for each  $s' \in \mathbb{S}$ :
        update  $V(s)$ 
```

which amounts to iterating over all triples $(s, a, s') \in \mathbb{S} \times \mathbb{A} \times \mathbb{S}$:

```
value_iteration:
  for each  $(s, a, s') \in \mathbb{S} \times \mathbb{A} \times \mathbb{S}$ :
    update  $V(s)$ 
```

where the update step runs in constant time. There are $\text{card}(\mathbb{S} \times \mathbb{A} \times \mathbb{S}) = |\mathbb{S}|^2|\mathbb{A}|$ loop iterations, each in constant time. The complexity of one iteration is therefore:

$$\Theta\left(|\mathbb{S}|^2|\mathbb{A}|\right)$$

and thus, performing n iterations (each constant-time **with respect to** n), the total complexity of VALUE ITERATION is:

$$\Theta\left(n|\mathbb{S}|^2|\mathbb{A}|\right)$$